A Terrain-Aware Full-Field Layout Algorithm and Receiver-Flux Benchmark for the Dunhuang 100 MW Solar Tower

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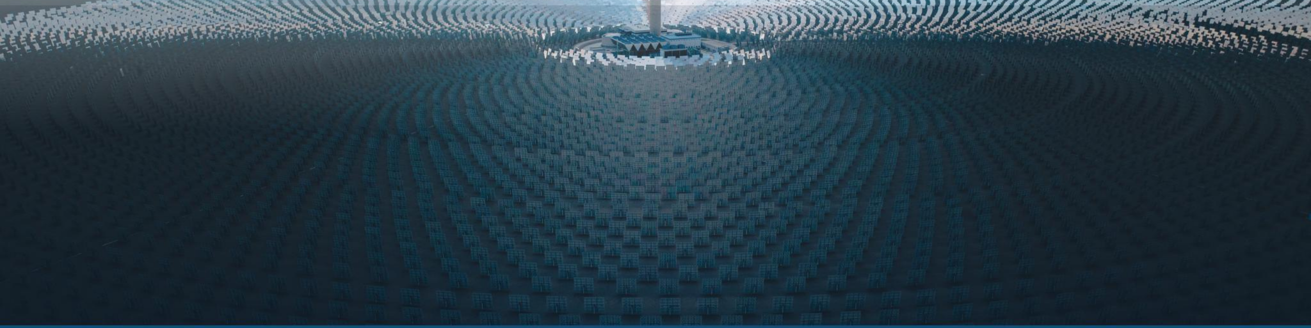


From audited coordinates to a receiver-risk-aware candidate queue

1. Full-field benchmark

with audited plant anchors

Plant-scale anchors

11915 coordinates retained

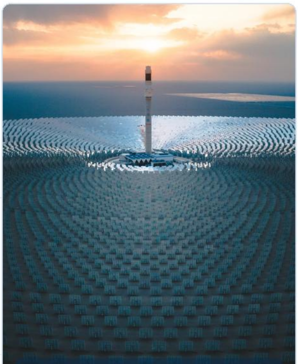
1.381M m2 mirrors

1883 annual DNI

~0.8% site slope constraint

- **Full-field** 11,915 coordinates
- **TS-FPDA** terrain-aware search
- **Direct gate** PySolTrace checked
- **Risk-aware** receiver-flux checked

Benchmark, not a claimed plant retrofit



Author aerial view, qualitative context

2. Algorithmic story

A real surround-field benchmark is turned into a bounded candidate queue.

01 **Audit** plant anchors + field count

02 **Generate** bounded full-field TFPDA variants

03 **Validate** SolarPILOT optical and default flux

04 **Re-rank** direct aimpoint PySolTrace queue

receiver-risk validation reorders the queue

What we deliver

- **Reproducible full-field layout algorithm**
- **Multi-layer validation package**
- **Receiver-flux-aware candidate queue**

3. Cross-layer evidence

Optical upper case **+3.15%**
D0276

efficiency gain under optical screening

Default flux penalty **+4.69%**
D0276

receiver loading flags the trade-off

Direct aimpoint check **-3.41%**
D1387

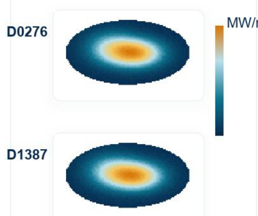
PySolTrace reorders receiver risk

Control role: D1822 remains conservative (-2.37% direct P/M).

4. Representative validation snapshots

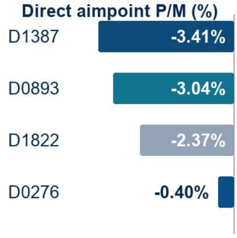
Optical gain vs. receiver penalty

Daily flux maps



Receiver-risk contrast

Direct aimpoint P/M (%)



PySolTrace queue reorders risk

Delta vs. default P/M (%)

5. Validation workload beyond the plant anchors

1841 candidates screened	**8** layouts exported	**27** solar conditions	**11** aiming strategies	**60000** ray hits per case	**1485** validation cases

This study contributes a reproducible full-field layout algorithm, validation package, and receiver-flux-aware candidate queue-not a final optimized Dunhuang redesign.

MATERIALS

Code, data tables, and validation package available with the article